



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch : IV Semester B.Tech IT
Course Code & Title : CS3451 & Introduction to Operating Systems
Name of the Faculty member : Mr. Balaganesh M V, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit 2 / CPU Scheduling
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO 5
- **Activity Chosen:** Roleplay
- **Justification:**

Students are asked to “act out” so they get a better idea of the concepts and theories being discussed. Role play encourages participation among passive learners, adds dynamism to the classroom and promotes the retention of material. CPU Scheduling is a process of determining which process will own CPU for execution while another process is on hold. The main task of CPU scheduling is to make sure that whenever the CPU remains idle, the OS at least select one of the processes available in the ready queue for execution.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - ❖ Instructor explained the particular concepts/topic in the classroom within 35 minutes.
 - ❖ The students formed groups and prepared for the role play.
 - ❖ The students enacted Round robin CPU scheduling algorithm.
 - ❖ Six processes with different burst time and arrival time are to be executed by the processor using round robin scheduling.
 - ❖ Students have taken the role of processes and processor as shown in fig 1&2.

• CO – PO / PSO mapping specific to this event:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO2	2	2	1	2	1	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	2	2	1	2	1	2
Justification for correlation	It involves logical thinking and make the students able to think about problem solving approach	Students will be able to understand the problem in real manner than theoretical explanation	Students will be able to provide solution for the problem	As the students are involved in the activity as team of two, the team work skill improved	Students communication skill will be improved as they are facing audience and communicating the activity	The problem solving skill earned through this activity helps the students in solving various problems

• **Images / Screenshot of the practice:**



Fig 1: Students involvement in CPU Scheduling as Processes (Banu Mathavan V, Dhinesh K, Dinesh Pandian G, Mohamed Islaam K A) Fig 2: Students involvement in CPU Scheduling as Processors (Sandhiya M, Ashmitha K V, Ananthi P, Abinaya S)

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Based on the activity and student's feedback, they felt that it was easy to understand the CPU scheduling concepts.
- Most of the students were actively participated in Roleplay rather than one-way communication.

❖ **Benefit of the practice:**

- Students understanding of the problem is improved.
- Students engage in higher order thinking and learn content in a deeper way.
- Students immediately apply content in a relevant, realistic scenarios.

❖ ***Challenges faced in implementation:***

- Making student participation was challenging. Initially, the students felt shy to participate
- Some of the students were discussing while playing.

References:

- ❖ <https://serc.carleton.edu/introgeo/roleplaying/whatis.html>

Signature of Faculty Member

HOD